Intergenerational Transmission of Trauma via Irritability: The case of 9/11 Responders with PTSD and their Adult Children

Daniel Craft[1], Diana V. Rodriguez-Moreno[1,2], Keely Cheslack-Postava[1,2], Aytan Waxman[1], Yael M. Cycowicz[1,2]

[1]New York State Psychiatric Institute, New York, New York; [2]Department of Psychiatry, Columbia University, New York, New York

Background

- Parental Post Traumatic Stress Disorder (PTSD) may lead to adverse mental health outcomes in children, a phenomenon referred to as Intergenerational Transmission of Trauma (ITT) [1-4]
- There are three proposed mediating mechanisms of ITT: parent-child role reversal, overprotective parenting, and parental anger/abuse [3,5]
- Previous research in World Trade Center (WTC) responders has found high rates of PTSD among responders [6], and increased rates of behavioral health issues among their children [7]
- We investigated ITT in WTC responders with a history of PTSD and their now-adult children by assessing their mental health and their perception of the relationship dynamic between them

Aims

To identify ITT pathways by examining associations between parental psychopathology and offspring's perception of the relationship, as well as offspring mental health outcomes.

Methods

Sample:

- 173 parent-child dyads were drawn from a larger ITT study
- Now-adult children were 19 - 41 years old; parents were 45 – 76 years old
- Responder parents were predominately male (86.13%), while more children were female (66.67%)

Measures:

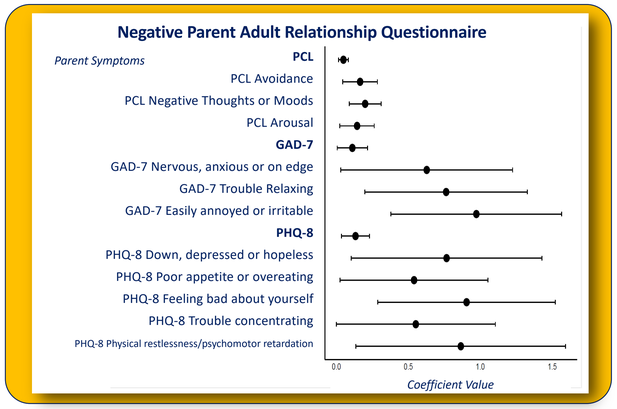
- Parent Adult Relationship Questionnaire (PARQ): 8-item survey measuring the perceived quality of the relationship between adults and their parent [8]
- Psychopathology was measured using standardized instruments:
  - Depression - Patient Health Questionnaire (PHQ-8) [9]
  - General Anxiety Disorder (GAD-7) [10]
  - PTSD Checklist V (PCL-V) [11]

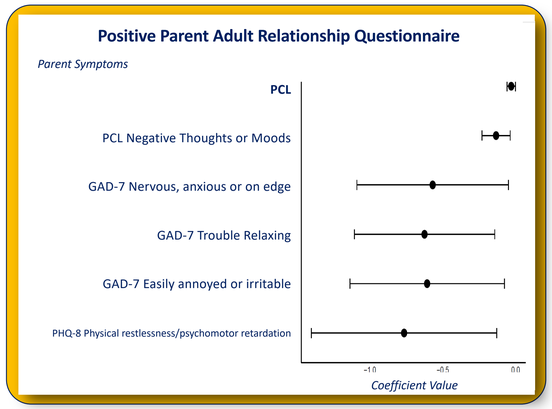
Analysis:

- We used PHQ-8, GAD-7 and PCL-V scores, as well as symptom cluster for PCL-V and individual symptoms for GAD-7 and PHQ-8. We used the Positive and Negative subscales and individual questions of the PARQ
- Regression coefficients were computed for three relations:
  - Parent Psychopathology and Child Responses on the PARQ
  - Child Psychopathology and Child Responses on the PARQ
  - Child Responses on Individual PARQ and Parent Symptoms

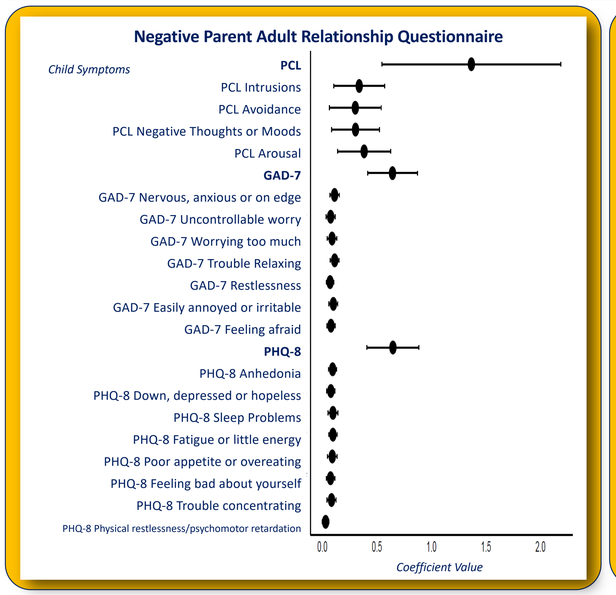
Results

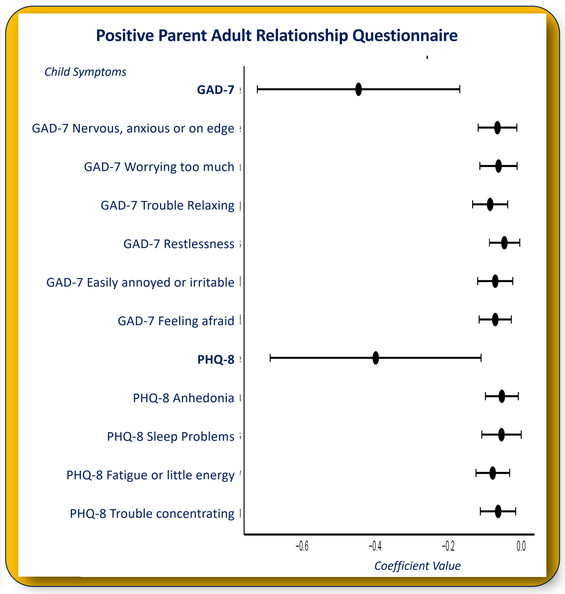
Analysis 1: Parent Psychopathology and Child Responses on the PARQ





Analysis 2: Child Psychopathology and Child Responses on the PARQ





Analysis 3:Child Responses on Individual PARQ and Parent Symptoms

Overall PCL

Child PARQ Question	β	SE	p
PARQ 2: Considerate	-0.009	0.004	0.031
PARQ 3: Favors	-0.010	0.004	0.026
PARQ 5: Angry or Hostile	0.016	0.005	0.002
PARQ 6: Unsympathetic	0.013	0.006	0.017
PARQ 7: Demanding	0.012	0.006	0.039

Overall GAD 7

	β	SE	p
PARQ 1: Affectionate	-0.033	0.015	0.029
PARQ 5: Angry or Hostile	0.043	0.015	0.006

Overall PHQ 8

	β	SE	p
PARQ 5: Angry or Hostile	0.043	0.014	0.003
PARQ 6: Unsympathetic	0.037	0.016	0.022
PARQ 8: Doubted	0.028	0.013	0.038

*Only statistically significant regression coefficients (β) shown

Cluster B: Intrusions

	β	SE	p
PARQ 5: Angry or Hostile	0.036	0.017	0.037

Cluster C: Avoidance

	β	SE	p
PARQ 4: Supportive	-0.033	0.015	0.023
PARQ 5: Angry or Hostile	0.052	0.018	0.004
PARQ 6: Unsympathetic	0.052	0.020	0.010

Cluster D: Negative Thoughts and Moods

	β	SE	p
PARQ 1: Affectionate	-0.042	0.017	0.014
PARQ 2: Considerate	-0.037	0.014	0.011
PARQ 3: Favors	-0.037	0.014	0.011
PARQ 4: Supportive	-0.034	0.014	0.015
PARQ 5: Angry or Hostile	0.054	0.017	0.001
PARQ 6: Unsympathetic	0.057	0.018	0.002
PARQ 7: Demanding	0.061	0.020	0.003

Cluster E: Arousal

	β	SE	p
PARQ 5: Angry or Hostile	0.050	0.018	0.005

*Only statistically significant regression coefficients (β) shown

Summary

- Greater parental psychopathology was consistently associated with perception of the parent as "angry or hostile"
- The parental symptom most strongly correlated with child negative PARQ score was "becoming easily annoyed"
- Negative perceptions of the parents were correlated with child PTSD symptomology, suggesting that ITT induced a vulnerability to PTSD for future trauma, with parent irritability as a key factor

Discussion

- Parental PTSD was related to increased vulnerability in children to developing PTSD, anxiety and depression symptoms
- The results support the parental abuse or anger mechanism, suggesting that the parents' struggles with trauma may lead to emotional outbursts or irritability, which their children perceived as aggression
- Other mechanisms were not supported in our study
- Parenting interventions for trauma-exposed individuals may reduce mental health risks for their children and strengthen family bonds

References

1. Rakoff V. (1966) A long term effect of the concentration camp experience. *Viewpoints* 1:17-22.
2. Lehrner, Amy, Bierer, Linda M, Passarelli, Vincent, Pratchett, Laura C, Flory, Janine D, Bader, Heather N, et al. (2014). Maternal PTSD associates with greater glucocorticoid sensitivity in offspring of Holocaust survivors. *Psychoneuroendocrinology*, *40*, 213-220. https://doi.org/10.1016/j.psyneuen.2013.11.019
3. Yehuda, R., & Lehrner, A. (2018). Intergenerational transmission of trauma effects: putative role of epigenetic mechanisms. *World Psychiatry*, 17(3), 243-257.
4. Yehuda, R., Halligan, S. L., & Grossman, R. (2001). Childhood trauma and risk for PTSD: Relationship to intergenerational effects of trauma, parental PTSD, and cortisol excretion. *Development and Psychopathology*, 13, 733–753.
5. Bar-On, D., Eland, J., Kleber, R.J., Krell, R.D., Moore, Y., Sagi, A., Soriano, E., Suedfeld, P., van der Velden, P.G., & van IJzendoorn, M.H. (1998). Multigenerational Perspectives on Coping with the Holocaust Experience: An Attachment Perspective for Understanding the Developmental Sequelae of Trauma across Generations. *International Journal of Behavioral Development, 22*, 315 - 338.
6. Soo J, Webber MP, Gustave J, et al. (2011). Trends in probable PTSD in firefighters exposed to the world trade center disaster, 2001-2010. *Disaster Med Public Health Prep*.;2:S197–S203.
7. Uchida, Mai, Huifen Feng, Adriana Feder, Natalie Mota, Clyde B. Schechter, Hilary D. Woodworth, Caroline G. Kelberman et al. (2018) "Parental posttraumatic stress and child behavioral problems in world trade center responders." *American Journal of Industrial Medicine* 61, no. 6: 504-514.
8. Pitzer, L., Fingerman, K., Lefkowitz, E. (2011). Development of the Parent Adult Relationship Questionnaire. *International Journal of Aging and Human Development*, 72 (2):111-135.
9. Kroenke K, Strine TW, Spritzer RL, Williams JB, Berry JT, Mokdad AH. (2009). The PHQ-8 as a measure of current depression in the general population. *Journal of Affect Disorders*. 114(1-3):163-73
10. Weathers, F. W., Litz, B. T., Keane, T. M., Palmieri, P. A., Marx, B. P., & Schnurr, P. P. (2013). The PTSD Checklist for DSM-5 (PCL-5) – Standard [Measurement instrument]. Available from https://www.ptsd.va.gov/
11. Spitzer, R. L., Kroenke, K., Williams, J. B., & Löwe, B. (2006). A brief measure for assessing generalized anxiety disorder: the GAD-7. *Archives of internal medicine*, 166(10), 1092-1097.

Acknowledgments

NIOSH/CDC U01OH012065, World Trade Center Health Program (WTCHP) and General Responder Data Center (GRDC).

Contact: daniel.craft@nyspi.columbia.edu